

Challenge (Habanero)

- Q1.** Factor the expression: $x^2 + 5x + 4x + 20$
(A) $(x + 5)(x + 4)$
(B) $(x + 4)(x + 5)$
(C) $(x + 2)(x + 10)$
(D) $(x + 1)(x + 20)$
(E) $(x + 3)(x + 12)$
- Q2.** Factor the expression: $3x^2 + 12x + 4x + 16$
(A) $3(x + 4)(x + 2)$
(B) $(3x + 4)(x + 4)$
(C) $(3x + 2)(x + 4)$
(D) $3(x + 2)(x + 4)$
(E) $(x + 2)(3x + 16)$
- Q3.** A gamer has x points in a game. After completing a level, the points are increased by $2x + 5x + 10$. Factor the expression $2x + 5x + 10$.
(A) $(2x + 5)(x + 2)$
(B) $(x + 2)(2x + 5)$
(C) $(x + 1)(7)$
(D) $(7)(x + 1)$
(E) $(7x + 10)$
- Q4.** A coding error causes a program to use $3x^2 + 6x + 2x + 4$ bytes of memory. Factor the expression.
(A) $3(x^2 + 2x) + 2(x + 2)$
(B) $3(x^2 + 2x + 1) + 2(x + 1)$
(C) $3(x + 1)(x + 2)$
(D) $3(x^2 + 2x) + 2(x + 2)$
(E) $3(x^2 + 2x + 2) + 2(x + 1)$
- Q5.** A data analyst is tracking website traffic. The total hits are modeled by $x^2 + 4x + 3x + 12$. Factor the expression.
(A) $(x^2 + 4x) + (3x + 12)$
(B) $(x + 3)(x + 4)$
(C) $(x + 4)(x + 3)$
(D) $(x^2 + 7x + 12)$
(E) $(x^2 + 3x + 12)$
- Q6.** A new update for a game uses $2x^2 + 8x + 3x + 12$ bytes of memory. Factor the expression.
(A) $2(x^2 + 4x) + 3(x + 4)$
(B) $2(x^2 + 4x + 1) + 3(x + 3)$
(C) $2(x + 4)(x + 1) + 3(x + 3)$
(D) $2(x^2 + 4x + 3) + 3(x + 1)$
(E) $2(x + 3)(x + 2)$
- Q7.** The scoreboard of a game is updated by $x^2 + 3x + 2x + 6$. Factor the expression.
(A) $(x + 3)(x + 2)$
(B) $(x + 2)(x + 3)$
(C) $(x^2 + 3x) + 2(x + 3)$
(D) $(x + 1)(x + 6)$
(E) $(x^2 + 5x + 6)$
- Q8.** A new level in a game has a data usage of $4x^2 + 12x + 3x + 9$. Factor the expression.
(A) $4(x^2 + 3x) + 3(x + 3)$
(B) $4(x^2 + 3x + 1) + 3(x + 2)$
(C) $4(x + 3)(x + 1)$
(D) $4(x^2 + 3x + 2) + 3(x + 1)$
(E) $(4x^2 + 12x + 9)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $2x^2 + 5x + 2x + 10$ and verify the result by multiplication.
- Q10.** The score of a game can be modeled by the expression $x^2 + 4x + 3x + 12$. Factor this expression and show that the result is correct by multiplying the factors.

Chef's Tips

- Use online resources to support student learning
- Encourage students to work in pairs to solve problems
- Provide feedback that guides students toward the correct solution